2026 INTERNATIONAL FINTECH COMPETITION / TOPIC C

# Technical Documentation

## Architecture, core algorithms, data provenance and evaluation

Technical explanation of EduRoute decision logic, the Digital Twin, Decision Passport, grounded AI, reference FX API, 30 directional country combinations, and sandbox transaction controls.

StudentBridge AI · China · United Kingdom · Japan · South Korea · Australia · Bangladesh

Prepared 15 October 2026 · Demonstration system · Real references with conditional user-input quotes

# Technical Architecture and Algorithm Description

## 1. Problem

An international student faces tuition amounts, multiple payment currencies, bank FX spreads, transfer charges, deadlines, incomplete country-pair eligibility data, unpredictable living expenses and unfamiliar destination credit systems. Many general-purpose converters do not jointly address these needs.

## 2. Objective

Provide explainable **decision support** for any of 30 ordered country pairs formed by China, UK, Japan, South Korea, Australia and Bangladesh, plus a safe, approved simulated execution as proof of task completion. No claim is made that all 30 real-money corridors are available.

## 3. Architecture

**Presentation layer:** One responsive HTML/CSS/JS dashboard with Planner, Digital Twin, Credit Navigator, Source Registry, Decision Passport, Sandbox Ledger, Grounded Copilot.

**API layer:** Python FastAPI and Pydantic schema validation. A successful live reference-rate request also returns a time-limited HMAC proof; the planner verifies that proof server-side before labeling a user-transmitted rate as truly API-backed. Important endpoints: `GET /api/catalog`, `GET /api/fx`, `POST /api/plans`, `GET /api/passport/{id}`, `POST /api/scenarios`, `POST /api/sandbox/execute`, `POST /api/invoices/analyze`, `GET /api/credit/{country}`, `POST /api/chat`, `GET /api/audit`.

**Deterministic finance layer:** `app/logic.py` uses `Decimal` arithmetic for core calculations, quantizes to 2 decimal places for display, checks hard safety constraints before ranking.

**Grounded agent layer:** Intent-based function calls return factual output. Optional Gemini integration only rephrases deterministic context; the LLM cannot call transaction execution.

**Storage layer: MySQL 8.0 database** for plans, approval-based simulated transactions, HMAC-chained audit events and all six research tables. SQLAlchemy + PyMySQL give parameterized statements; plan/transaction + audit writes are atomic. HMAC chain updates lock the MySQL chain-head row. Read-only source registry/academic catalog is JSON with provenance and source dates.

## 4. EduRoute algorithm

Let `T` be the invoice tuition in destination currency and `S` the confirmed scholarship reduction. Required tuition `R=max(T-S,0)`.

For a provider quote `q`, with provider sell rate `r` (origin currency per one destination currency), percentage fee `p` and fixed fee `f` in origin currency, base cost `B=R×r`, fee `F=B×p/100+f`, total `C=B+F`.

`deadline_ok = quoted_eta_days <= remaining_days` (only an estimate). `affordable = C <= available_funds - protected_reserve`. `eligible = student_acknowledged_provider_eligibility`. Quotes missing a source URL are flagged (not silently treated as verified). Expired quotes are rejected. A viable option requires eligibility, deadline and affordability, and no expiry error. Among viable quotes, rank primarily by ascending total cost and secondarily by arrival estimate.

**Important:** `eligible` is a *user assertion*, not an automated regulator/provider determination; therefore the output remains a **conditional quote ranking**, never a verified real-world bank recommendation.

When no quote is viable, return a refusal to recommend with an explanation. A real reference FX observation without fees never generates a provider quote.

## 5. Financial Digital Twin

Apply a user-selected percentage shock `s` to the quoted conversion base: `B' = B × (1+s/100)`. Recalculate percentage transfer fee, add fixed fee plus user-selected incremental costs, compare the result to baseline. Check whether protected reserve remains intact. Recompute destination living-cost budget separately with a user-selected living-cost shock.

The simulation is a counterfactual what-if calculator, **not an FX forecast** or probabilistic claim. User-specified scenarios are hypothetical even if the base price comes from a real quote.

## 6. Decision Passport

The passport contains a plan identifier, created timestamp, goal, input constraints, each quote's calculated cost and rejection reasons, recommendation, reference FX provenance and caution labels. Every plan includes a SHA-256 integrity fingerprint of the deterministic result before the ID and fingerprint are attached. This is a local integrity indicator, not third-party notarization.

## 7. Invoice document analysis

The invoice endpoint processes only PDF bytes (`%PDF` signature), limited to 2.5 MB and 12 pages. PyMuPDF extracts text; conservative regular expressions surface **candidate** amounts and ISO-formatted dates. The student confirms these inputs manually. File text is never run as code or agent instructions. No recipient/bank account verification, OCR or invoice-fraud certification is claimed.

## 8. Agent reasoning and tool decisions

The assistant selects a trusted tool path based on task intent. Credit questions retrieve country guidance with primary references. Route questions return the deterministic planner decision and label quotes as user entered. Budget questions report entered funds and availability. Tuition questions select official published tuition records when available. Optional Gemini makes the text easier to read but does not create numbers or authorization decisions. There is no external agent execution autonomy in this MVP.

## 9. Testing

Automated tests exercise core arithmetic, scholarships, deadline constraints, eligibility assertions, emergency reserve, expired quotes, no-offer response, FX stress shocks, wrong currencies, explicit approval, duplicate idempotency, invoice type checks, country coverage and grounded tool selection. Run `python -m pytest -q`.

## 10. Improvements after the competition

Commercial provider quotation APIs with authenticated quote/eligibility/expiry, payment corridor rules reviewed by local counsel, actual student invoice beneficiary verification, user authentication, HTTPS deployment, encryption-at-rest, rate limiting, secrets management, signed audit attestations, accessibility audit, locale/currency rounding rules, price snapshotting, OCR, production observability, rigorous bias and robustness evaluation. None is represented as complete in the current package.

## 11. MySQL 8 deployment design (15 October 2026)

Persistent application and research data are stored in ten MySQL tables. The schema is generated as `scripts/mysql_schema.sql`; corrected official research CSV rows can be safely imported with `scripts/setup_mysql.py`. Foreign keys are used for plans/transactions, research sources, country codes and source metadata. MySQL InnoDB provides ACID transactions and row locking for serializing chained audit entries. Each sandbox transaction and its audit event use a single database transaction. Idempotency keys have a database-level UNIQUE constraint to prevent duplicate writes. Missing provider quotes remain NULL and are not made actionable. No real banking integration or real student records are included. Automated unit tests run in disposable isolated SQLite only and live MySQL smoke verification is a separate step; user must test actual MySQL deployment.

# Real Financial Data Collection - Exact Instructions

## Scope and data status

Support country selection for China (CNY), United Kingdom (GBP), Japan (JPY), South Korea (KRW), Australia (AUD) and Bangladesh (BDT). There are 30 ordered directions, not 30 proven bank routes. All data must include source URLs, observation/effective dates and `source_type`. Missing data stays missing. Real student financial information requires consent, minimum necessary fields and secure treatment; none is bundled.

## 1. Live reference FX (six currencies)

**Source:** https://frankfurter.dev/ (v2). API example: `GET https://api.frankfurter.dev/v2/rate/JPY/GBP` returns a numerical rate and publication date, when available. The backend fetches only on demand; if internet is unreachable, it returns `available=false` with no invented rate.

**Fields:** `base_currency, quote_currency, published_rate, observation_date, retrieved_at, source_url, rate_type=market_reference`.

**Never:** call it a provider sell rate or guarantee that customers can transfer at that price.

## 2. Real 2026 university tuition

**Sources:**

- Australia (specific numeric data): https://students.unimelb.edu.au/__data/assets/pdf_file/0020/5423033/2026-International-Course-Tuition-Fees-OF_20260507.pdf
- United Kingdom: https://www.manchester.ac.uk/study/masters/courses/list//13005/ma-international-education/overview/ and https://www.ucl.ac.uk/prospective-students/graduate/taught-degrees/management-complex-projects-msc
- China: https://masters.sem.tsinghua.edu.cn/en/info/1007/2103.htm
- Japan: https://ipads.a.u-tokyo.ac.jp/admissions/cost-scholarship/ and https://www.s.u-tokyo.ac.jp/en/admission/research-students.html
- South Korea: https://ee.kaist.ac.kr/en/admission-08/
- Bangladesh: https://www.bracu.ac.bd/admissions/tuition-and-fees (per-credit published data; individual fee status matters).

**Fields:** `institution, country, programme, year, student_fee_status, tuition_amount, currency, payment_period, campus, source_url, updated_at, exact_invoice_confirmed`.

University-published annual/cost-per-EFTSL data must never be automatically confused with the student's actual payable invoice. The catalog now includes selected 2026 official published fee examples for all six countries; it always shows period, source and any student-status restrictions.

## 3. Bank and cross-border provider offers

**Source:** actual bank/provider customer quotation applicable to **the exact two countries and currency direction**. Ask the bank for documented tuition-purpose eligibility, payment method, signed/net beneficiary amount, quoted rate, flat fees, percentage fees, intermediary fees, cut-off time, expected settlement, quote expiry and beneficiary validation rules. A general price page is *not* a customer-specific quote.

**Fields:** `provider, source_country, destination_country, corridor_currency_pair, declared_eligibility, quote_rate, fixed_fee, percent_fee, correspondent_fees, beneficiary_deductions, expected_eta, quote_timestamp, expiry, reference_url_or_document_id, verification_status`.

**If missing:** `quote_unavailable`. Do not generate fictional "real" bank offers. Some information may require a licensed API, bank relationship, or logged-in customer account.

## 4. Regulatory references

Start from the official national regulators: China SAFE (`safe.gov.cn/en`), UK FCA (`fca.org.uk`), Japan FSA (`fsa.go.jp/en`), Korea FSC (`fsc.go.kr/eng/index`), Australia AUSTRAC (`austrac.gov.au`), Bangladesh Bank (`bb.org.bd`). These are *research starting points* rather than machine-readable verified eligibility for each route.

**Fields:** `origin, destination, type, relevant_entity, law_or_rule, allowed_conditions, document_checklist, applicable_thresholds, effective_date, source_url, reviewed_by, legal_review_status`.

The current system cannot assert that a cross-border transfer is legally compliant. Do not replace proper KYC/AML with a source URL.

## 5. Living costs and formal visa funds

**Real examples:** UK Student visa London GBP 1,529 per month / outside London GBP 1,171 per month up to 9 months, with applicable exceptions, from https://www.gov.uk/student-visa/money. JASSO Student Guide 2025-26: 2023 privately financed international student living costs JPY 105,000 monthly national average; source https://www.studyinjapan.go.jp/en/assets/pdf/sgtj_2025_english.pdf.

**Fields:** `country, city, amount, currency, month, category, scope, observation_year, publication_date, source_url, is_visa_rule`. Do not present visa bank-balance requirements as a guarantee of actual living affordability.

## 6. Credit readiness

Official/principal examples: UK MoneyHelper (`moneyhelper.org.uk`), Australian Government Moneysmart (`moneysmart.gov.au`), Japan CIC (`cic.co.jp/en/credit-report/`). Add China, South Korea and Bangladesh details only from verified local institutions. Store `country, consumer_action, document_requirement, eligibility_caveat, official_url, effective_date`.

Credit score prediction is out of scope. Do not invent students' private credit profiles.

## 7. Student-provided information

The user can enter destination university, confirmed tuition amount and deadline, scholarship already awarded, actual *self-reported* available money, protected reserve and living-expense estimates. The project does not request passwords, card PIN, OTP, government identifier, passport scan or real bank credentials. The sandbox DB must not be treated as secure storage for sensitive real data.

## 8. Practical collection order

**Day 1:** load FX API reference and 6-country catalog; collect a narrow set of university fees with original links. **Day 2:** enter bank/provider quotes for actual supported routes, explicitly recording gaps and confirming expiry. **Day 3:** map tuition document conditions and visa/living references. **Day 4:** review student credit guidance and build additional institution coverage. **Day 5:** revalidate source timestamps and run financial/error tests.

## 9. Provenance checklist

For each record ask: Is this a *primary* source? Is it applicable to 2026 and the intended customer? Is the value a price, an estimate, a statutory requirement, or a quote? Does it specify which way the currency converts? Is it

available for the requested pair? Has it expired? Is any charge missing? Are there redistribution/licensing restrictions? If any answer is unknown, display a verification-needed label.

# Competition Rules Compliance Matrix (2026 FinTechathon International Track)

This project follows the user's uploaded **PBCSF Appendix 2 - Problem Description and Competition Rules for International Track** (4 pages), included at `docs/Competition_Rules.pdf`.

## Topic C

Official topic: "Cross-border & Student Finance Assistant" - remittances, tuition payments, credit building and multi-currency management. StudentBridge AI supports those use cases as follows: cross-border tuition planning with a quote-aware optimized ranking; multiple currencies and historical reference FX API; personal financial digital twin; credit-readiness education; secure sandbox execution.

## Judging weights

- **Task completion 40%:** end-to-end user inputs -> decision plan -> risk check -> user approval -> test transaction -> receipt -> audit event.
- **Security & compliance 30%:** controls and explicitly documented limitations. Does not claim KYC/AML screening or regulatory certification; blocking unsafe inputs and approval checks can be demonstrated.
- **Innovation & interaction 30%:** deadline-constrained optimizer, reserve-aware Digital Twin, source-linked Decision Passport, six-country choice, responsive dashboard and grounded copilot.

## Submission materials required (page 3)

1. Technical documentation - provided as system design `.md` and `.pdf`.
2. Presentation - `docs/submission/StudentBridge_AI_Presentation.pptx`. Keep spoken presentation under 10 minutes.
3. Demo video - optional standalone demo, max 5 minutes; not recorded in package.
4. Source code, README, repository link, deployment directions, test cases - all provided locally except live GitHub link. **Publish to GitHub yourself**.
5. Security self-assessment - in PDF and Markdown, with permission tiers and risks.
6. Execution evidence for non-on-chain topic - sample simulated-account ledger and logs. A real **screen demonstration video still needs recording and submission**.

## Administrative requirements (page 3-4)

- Teams of 2-5 students; 60% or more non-Chinese nationals for the International Track, and eligible international institutions.

- English for presentations, submissions and reviews.
- October 20, 2026 23:59 Beijing time (October 20, 2026 21:59 Bangladesh time) deadline.
- Only one team per participant; independent original work, properly licensed sources, and no plagiarism.
- Finalists: at least one member must attend in person or risk forfeiture of awards.

The submission package **is not automatically submitted** and does not imply the student's eligibility or source-license review has been completed.